

Doom PWAD QA Report

Run 4d96faa0-be56-4871-8ce8-a604f743f1d8 | Map E1M1 | Outcome stuck

Executive Summary

Document the automated PWAD QA run, combine static map analysis with live play telemetry, and give map authors concrete evidence about completion, traversal, combat, and defects. The selected-difficulty gameplay evidence uses 0 spawned enemy/enemies. Raw static analysis found 8 enemies, but 8 are disabled at difficulty 1 by skill flags.

Pass/fail:

map_navigation: FAIL

combat_engagement: PASS

resource_balance: PASS

secret_coverage: PASS

overall_verdict: FAIL

- The run ended with outcome 'stuck' after visiting 6 coarse position cluster(s).
- 0 kill(s) recorded against 0 enemy/enemies that spawn at difficulty 1.
- Selected-difficulty health_ratio=0.0000, ammo_ratio=0.0000.
- No secret sectors exist in static analysis.

The highest-severity open finding is 'Run stalled after repeated stuck decisions' at severity 1. Review this before publishing the map.

Test Item And Environment

Test item: E1M1 from WAD ac683d7a-65e6-4b06-b94f-b7c0cf8961af. Static analysis context: 10 things, 8 enemies, 1 items, 0 keys, 0 secret sectors, 35 linedefs, 5 sectors, estimated difficulty hard. At difficulty 1, 0 of 8 raw enemies spawn.

The run used IWAD freedoom1, difficulty 1, model gemini-3.1-flash-lite, and a maximum budget of 900 game ticks.

Metric	Value	Metric	Value
Status	completed	Duration	19
Actions	13	LLM calls	13
Final HP	100	Kills	0
Position samples	233	Movement units	1698.7
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/4d96faa0-be56-4871-8ce8-a604f743f1d8.mp4		

Recording And Agent Quality

Recording	Value	Agent progress	Value
Status	ok	Progress score	0
Frames	95 total / 95 unique	Meaningful progress	0
Video shape	640 x 480 @ 15.0 FPS	Completed objects	0
Gameplay time	6.286s, 220 tic(s)	Quality flags	warning

Quality warnings:

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Static Analysis Context

Metric	Raw WAD	Difficulty 1	Notes
Enemies	8	0	8 hidden by skill/multiplayer flags
Items	1	0	1 hidden by skill/multiplayer flags
Secrets	0	0	Sector special 9
Geometry	35 linedefs	5 sectors	2528 x 1056 units

Selected-difficulty estimate: easy. Health ratio: 0.0. Ammo ratio: 0.0.

Coverage Evaluation

The automated playthrough recorded 13 gameplay event(s), 13 lockstep LLM/MCP decision(s), 233 movement samples, 6 position cluster(s), and 1698.7 map units of movement. The run ended with outcome 'stuck'. Combat coverage is evaluated against 0 enemy/enemies that spawn at difficulty 1, not against hidden raw THINGS. Recording quality status: ok; 95 frames, 95 unique frames, 6.333 seconds at 15.0 FPS. Automated playthrough progress score: 0 with 0 meaningful progress event(s). Quality flags: warning.

Objectives Covered

- **Map launch and initial state capture:** Run status: completed.
- **Director decision logging:** 13 decision event(s) recorded.
- **Movement telemetry:** 233 position sample(s) across 6 cluster(s).
- **Gameplay recording:** /media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/4d96faa0-be56-4871-8ce8-a604f743f1d8.mp4

Objectives Omitted

- **Verified map exit:** Run outcome was stuck.

Uncovered attributes: exit reliability, secret accessibility

Defects And Risks

2 defect(s) were detected. Primary risk: at least one detected defect needs author review.

Patterns: difficulty_spawn_mismatch: 1; softlock_navigation: 1

Severity	Priority	Type	Tick	Description
1	1	softlock_navigation	188	Run stalled after repeated stuck decisions The run produced repeated stuck events with no meaningful progress. This usually indicates blocked navigation, unreachable objectives, or an automated control loop that should be reviewed with the recording and action trace.

2	2	difficulty_spawn_mismatch	0	Difficulty-specific thing flags hide map content At difficulty 1, 8 of 8 enemies and 1 of 1 items from raw static analysis do not spawn in single-player because of Doom skill or multiplayer flags.
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Risk Areas

- **Progression:** Map exit was not verified; outcome was stuck.
- **Difficulty flags:** 8 enemy/enemies are present in raw THINGS but do not spawn at difficulty 1.
- **softlock_navigation:** Run stalled after repeated stuck decisions
- **difficulty_spawn_mismatch:** Difficulty-specific thing flags hide map content

Activity Log Summary

The pipeline statically analyzed the map, launched a ViZDoom MCP session, executed 13 lockstep MCP action(s), collected event and position telemetry, finalized recording artifacts, detected defects, and generated this report.

Run data is sparse or quality flags are present; treat gameplay conclusions as preliminary.

Limitations And Recommendations

8 raw enemy/enemies did not spawn at difficulty 1; combat conclusions only apply to spawned enemies. Automated play can miss secrets or optional combat paths that require map-specific intent. Single-run results should be confirmed with additional seeds or manual review before release decisions.

Good Quality Areas

- **Runtime stability:** The backend completed the run lifecycle.
- **Telemetry:** The run includes multiple position samples.
- **Evidence capture:** A gameplay recording artifact was produced.
- **Static context:** The report includes parsed map structure and thing counts.

Evidence Appendix

Notable Event Trace

Tick	Event	Position	HP	MCP Action	Stop reason
188	stuck	-16.8, 133.5	100	take_action	tics_complete
193	stuck	-12.6, 153.4	100	take_action	tics_complete
196	stuck	-10.7, 164.4	100	take_action	tics_complete
224	stuck	-19.5, 83.3	100	retreat	complete
226	stuck	-20.3, 76.9	100	take_action	tics_complete
231	stuck	-34.9, 40.5	100	take_action	tics_complete
234	stuck	-43.3, 26.1	100	take_action	tics_complete

LLM/MCP Decision Trace

#0 ticks 14 -> 94 tool explore stop max_tics Gemini API key is not configured; using deterministic fallback. No visible combat or pickup target is available, so a short bounded exploration step starts.
#1 ticks 95 -> 174 tool explore stop max_tics Gemini API key is not configured; using deterministic fallback. No visible combat or pickup target is available, so a short bounded exploration step starts.
#2 ticks 175 -> 177 tool take_action stop tics_complete Gemini API key is not configured; using deterministic fallback. Recent exploration did not progress, so probing a nearby door/use interaction.
#3 ticks 178 -> 180 tool take_action stop tics_complete Gemini API key is not configured; using deterministic fallback. Recent exploration did not progress, so probing a nearby door/use interaction.
#4 ticks 181 -> 183 tool take_action stop tics_complete Gemini API key is not configured; using deterministic fallback. Recent exploration did not progress, so probing a nearby door/use interaction.
#5 ticks 184 -> 186 tool take_action stop tics_complete Gemini API key is not configured; using deterministic fallback. Recent exploration did not progress, so probing a nearby door/use interaction.
#6 ticks 187 -> 188 tool take_action stop tics_complete The requested action repeats a recent no-progress signature (take_action), so I am switching tactics. I am facing a fresh unexplored direction first, then I will move in short bounded steps.
#7 ticks 189 -> 193 tool take_action stop tics_complete Progress has not changed across repeated lockstep decisions, so I am forcing a bounded QA recovery probe. I am advancing straight under direct control instead of letting explore arc in place.

#8 | ticks 194 -> 196 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (take_action), so I am switching tactics. I am probing for a switch or door interaction before declaring the area blocked.

#9 | ticks 197 -> 224 | tool retreat | stop complete

The requested action repeats a recent no-progress signature (take_action), so I am switching tactics. The direct probes did not progress yet, so I am retreating and rotating out of the loop.

#10 | ticks 225 -> 226 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (take_action), so I am switching tactics. I am facing a fresh unexplored direction first, then I will move in short bounded steps.

#11 | ticks 227 -> 231 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (take_action), so I am switching tactics. I am advancing straight under direct control instead of letting explore arc in place.

#12 | ticks 232 -> 234 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (take_action), so I am switching tactics. I am probing for a switch or door interaction before declaring the area blocked.